

SciNoNet: A Helmholtz-Potential Physics-Informed Network Architecture, Governing Physics, and Loss Functions

1 Overview

The model reconstructs the three displacement components u, v, w of guided elastic (Lamb) waves on an aluminum plate. A neural network predicts four *Helmholtz potentials* from the space-time coordinates; the displacement is obtained from the potentials by the Helmholtz decomposition, and the potentials are constrained to satisfy two scalar wave equations (one longitudinal, one shear), a gauge condition, and a rest initial condition. Every quantity is expressed in standardized (non-dimensional) variables; Section 2 defines them and Table 1 lists every symbol with its value.

2 Coordinates, fields, and parameters

The physical inputs are the position (x, y, z) in mm and time t in s; the outputs are the displacements (u, v, w) in mm. The plate is $300 \times 200 \times 2$ mm with the through-thickness direction z . The data are standardized to zero-centered, $O(1)$ variables. With sample means $\mu_\bullet$ and standard deviations as defined below,

$$\hat{x} = \frac{x - \mu_x}{L}, \quad \hat{y} = \frac{y - \mu_y}{L}, \quad \hat{z} = \frac{z - \mu_z}{L}, \quad \hat{t} = \frac{t - \mu_t}{s_t}, \quad (1)$$

where L is a single in-plane length scale used for *all three* spatial axes (including z), and s_t is the time scale. Using one length L for z as well (rather than the small plate thickness) keeps the spatial derivatives balanced; this is captured by the ratio

$$\rho \equiv \frac{L}{L_z}, \quad L_z = L \Rightarrow \rho = 1, \quad (2)$$

which multiplies every z -derivative term below; the equations are written for general ρ and reduce to the implemented case at $\rho = 1$. The fields are standardized by a single common scale s_f (their combined standard deviation):

$$\hat{u} = \frac{u}{s_f}, \quad \hat{v} = \frac{v}{s_f}, \quad \hat{w} = \frac{w}{s_f}. \quad (3)$$

3 Network architecture

The network maps the standardized coordinates $\mathbf{x} = (\hat{x}, \hat{y}, \hat{z}, \hat{t})$ to the four scalar potentials

$$\mathbf{p}(\mathbf{x}) = (\hat{\phi}, \hat{\psi}_x, \hat{\psi}_y, \hat{\psi}_z) = \text{MLP}(\gamma(\mathbf{x})), \quad (4)$$

where $\hat{\phi}$ is the (dimensionless) scalar potential and $\hat{\boldsymbol{\psi}} = (\hat{\psi}_x, \hat{\psi}_y, \hat{\psi}_z)$ the vector potential.

Symbol	Meaning	Value (this run)
μ_x, μ_y, μ_z	coordinate means	data-dependent
L	in-plane length scale (=std of x, y)	≈ 89 mm
L_z	through-thickness scale (= L)	≈ 89 mm
$\rho = L/L_z$	in-plane/thickness ratio	1
s_t	time scale (=std of t)	$\approx 1.73 \times 10^{-5}$ s
s_f	field scale (=std of u, v, w)	$\approx 6.7 \times 10^{-6}$ mm
c_p	longitudinal (P) wave speed	6.3×10^6 mm/s
c_s	shear (S) wave speed	3.15×10^6 mm/s
$\hat{c}_p = c_p s_t / L$	standardized P speed	≈ 1.22
$\hat{c}_s = c_s s_t / L$	standardized S speed	≈ 0.61
L_f	number of Fourier frequencies	160
α	gradient-balancing target ratio	0.3
w_g, w_i	gauge / IC sub-weights	1, 1
N_c, N_i	collocation / IC batch sizes	2048, 1024
Δt	simulation timestep	10^{-8} s

Table 1: Symbols and their values.

Fourier-feature embedding. The coordinates are first lifted by a specialized random-Fourier embedding. With a frequency matrix $\mathbf{B} \in \mathbb{R}^{4 \times L_f}$ (columns $\mathbf{B}_{:,j}$) and a per-frequency amplitude A_j ,

$$\gamma(\mathbf{x}) = \left[A_1 \sin(2\pi \mathbf{x} \cdot \mathbf{B}_{:,1}), \dots, A_{L_f} \sin(2\pi \mathbf{x} \cdot \mathbf{B}_{:,L_f}), A_1 \cos(\cdot), \dots, A_{L_f} \cos(\cdot), \mathbf{x} \right] \in \mathbb{R}^{2L_f+4}. \quad (5)$$

The raw coordinates $\mathbf{x}$ are appended (the trailing term) so the network can also represent slow, linear trends. Both A_j and $\mathbf{B}$ are learnable; $\mathbf{B}$ is initialized uniformly within and clamped after each step to per-dimension bands taken from the wave dispersion curve. In standardized cycles-per-unit (a physical frequency f maps to $f \cdot \text{std}$),

$$B_x, B_y, B_z \in [-k_{\max} L, k_{\max} L], \quad B_t \in [0, f_{\max} s_t], \quad (6)$$

with $k_{\max} = 0.26$ cyc/mm (= 2×0.13 , the band $\kappa \leq 0.8$ rad/mm) and $f_{\max} = 300$ kHz, giving $|B_{x,y,z}| \lesssim 23$ and $B_t \lesssim 5.2$.

MLP. The embedding feeds a multilayer perceptron with three hidden layers of width 256 and tanh activations, ending in a linear layer with four outputs (the potentials). The two wave speeds enter as fixed parameters $\hat{c}_p, \hat{c}_s$ (Section 5); they are stored as learnable log-parameters but held fixed here so that c_p/c_s stays at its physical value.

4 Displacement from the potentials

The Helmholtz decomposition writes the displacement field as the gradient of the scalar potential plus the curl of the vector potential, $\mathbf{u} = \nabla \Phi + \nabla \times \Psi$, i.e.

$$u = \frac{\partial \Phi}{\partial x} + \frac{\partial \Psi_z}{\partial y} - \frac{\partial \Psi_y}{\partial z}, \quad v = \frac{\partial \Phi}{\partial y} + \frac{\partial \Psi_x}{\partial z} - \frac{\partial \Psi_z}{\partial x}, \quad w = \frac{\partial \Phi}{\partial z} + \frac{\partial \Psi_y}{\partial x} - \frac{\partial \Psi_x}{\partial y}, \quad (7)$$

where Φ, Ψ are the *physical* potentials. They are related to the network’s dimensionless outputs by a single scale $C = s_f L$,

$$\Phi = C \hat{\phi}, \quad \Psi_x = C \hat{\psi}_x, \quad \Psi_y = C \hat{\psi}_y, \quad \Psi_z = C \hat{\psi}_z. \quad (8)$$

From (1), a physical derivative becomes a standardized one through the chain rule, $\partial/\partial x = (1/L)\partial/\partial\hat{x}$, $\partial/\partial z = (1/L_z)\partial/\partial\hat{z}$. Substituting into the first line of (7),

$$\begin{aligned} u &= C\left(\frac{1}{L}\hat{\phi}_{\hat{x}} + \frac{1}{L}\hat{\psi}_{z,\hat{y}} - \frac{1}{L_z}\hat{\psi}_{y,\hat{z}}\right) = \frac{C}{L}\left(\hat{\phi}_{\hat{x}} + \hat{\psi}_{z,\hat{y}} - \frac{L}{L_z}\hat{\psi}_{y,\hat{z}}\right) \\ &= s_f(\hat{\phi}_{\hat{x}} + \hat{\psi}_{z,\hat{y}} - \rho\hat{\psi}_{y,\hat{z}}), \end{aligned} \quad (9)$$

using $C/L = s_f$. Dividing by s_f gives the standardized displacement the data loss compares against (subscripts after a comma denote partial derivatives):

$$\boxed{\hat{u} = \hat{\phi}_{\hat{x}} + \hat{\psi}_{z,\hat{y}} - \rho\hat{\psi}_{y,\hat{z}}, \quad \hat{v} = \hat{\phi}_{\hat{y}} + \rho\hat{\psi}_{x,\hat{z}} - \hat{\psi}_{z,\hat{x}}, \quad \hat{w} = \rho\hat{\phi}_{\hat{z}} + \hat{\psi}_{y,\hat{x}} - \hat{\psi}_{x,\hat{y}}.} \quad (10)$$

All derivatives are obtained by automatic differentiation of the network output.

5 Governing physics

5.1 Wave equations and their standardized residual

For an isotropic elastic solid the scalar potential carries the longitudinal (P) wave at speed c_p and each component of the vector potential carries the shear (S) wave at speed c_s :

$$\frac{\partial^2 \Phi}{\partial t^2} = c_p^2 \nabla^2 \Phi, \quad \frac{\partial^2 \Psi_k}{\partial t^2} = c_s^2 \nabla^2 \Psi_k \quad (k = x, y, z), \quad (11)$$

with $\nabla^2 = \partial_{xx} + \partial_{yy} + \partial_{zz}$. The two fundamental Lamb modes (symmetric S_0 , antisymmetric A_0) arise from the superposition of Φ and Ψ through the plate thickness. We derive the residual for a generic potential $q \in \{\hat{\phi}, \hat{\psi}_x, \hat{\psi}_y, \hat{\psi}_z\}$ with speed $c \in \{c_p, c_s\}$. Writing the physical potential $Q = Cq$ and using the chain rule on (11),

$$\frac{\partial^2 Q}{\partial t^2} = \frac{C}{s_t^2} q_{\hat{t}\hat{t}}, \quad \nabla^2 Q = C\left[\frac{1}{L^2}(q_{\hat{x}\hat{x}} + q_{\hat{y}\hat{y}}) + \frac{1}{L_z^2}q_{\hat{z}\hat{z}}\right]. \quad (12)$$

Substituting and cancelling C ,

$$\frac{1}{s_t^2} q_{\hat{t}\hat{t}} = c^2 \left[\frac{1}{L^2}(q_{\hat{x}\hat{x}} + q_{\hat{y}\hat{y}}) + \frac{1}{L_z^2}q_{\hat{z}\hat{z}}\right] \implies q_{\hat{t}\hat{t}} = \hat{c}^2 (q_{\hat{x}\hat{x}} + q_{\hat{y}\hat{y}}) + \hat{c}^2 \rho^2 q_{\hat{z}\hat{z}}, \quad (13)$$

where $\hat{c} = c s_t/L$ is the standardized speed and we used $c^2 s_t^2/L^2 = \hat{c}^2$ and $c^2 s_t^2/L_z^2 = \hat{c}^2 \rho^2$ (since $1/L_z^2 = \rho^2/L^2$). Dividing through by $\hat{c}^2 \rho^2$ to keep all terms $O(1)$ gives the residual used in the loss,

$$\boxed{r_q = \frac{1}{\hat{c}^2 \rho^2} q_{\hat{t}\hat{t}} - \frac{1}{\rho^2} (q_{\hat{x}\hat{x}} + q_{\hat{y}\hat{y}}) - q_{\hat{z}\hat{z}},} \quad (14)$$

with $\hat{c} = \hat{c}_p$ for $q = \hat{\phi}$ and $\hat{c} = \hat{c}_s$ for the three $\hat{\psi}_k$. At $\rho = 1$ this is $r_q = q_{\hat{t}\hat{t}}/\hat{c}^2 - (q_{\hat{x}\hat{x}} + q_{\hat{y}\hat{y}} + q_{\hat{z}\hat{z}})$. A solution of the wave equation makes $r_q = 0$.

5.2 Coulomb gauge

The vector potential is not unique: $\Psi \rightarrow \Psi + \nabla\chi$ changes nothing in $\nabla \times \Psi$, hence nothing in $\mathbf{u}$. This freedom is removed by the Coulomb gauge $\nabla \cdot \Psi = 0$. In standardized variables,

$$\nabla \cdot \Psi = \frac{\partial \Psi_x}{\partial x} + \frac{\partial \Psi_y}{\partial y} + \frac{\partial \Psi_z}{\partial z} = \frac{C}{L}(\hat{\psi}_{x,\hat{x}} + \hat{\psi}_{y,\hat{y}} + \rho\hat{\psi}_{z,\hat{z}}), \quad (15)$$

so the (scale-free) gauge residual is

$$\boxed{g = \hat{\psi}_{x,\hat{x}} + \hat{\psi}_{y,\hat{y}} + \rho\hat{\psi}_{z,\hat{z}}.} \quad (16)$$

5.3 Initial condition

The plate is at rest before excitation: at $t = 0$ the displacement and velocity vanish, which (with the gauge) is enforced by requiring the potentials and their time-derivatives to vanish on the $t = 0$ plane,

$$\mathbf{p}|_{\hat{t}_0} = \mathbf{0}, \quad \frac{\partial \mathbf{p}}{\partial \hat{t}}|_{\hat{t}_0} = \mathbf{0}, \quad \hat{t}_0 = \frac{0 - \mu_t}{s_t}. \quad (17)$$

The eight scalar residuals are stacked into $\mathbf{r}_{\text{ic}} = (\mathbf{p}, \partial \mathbf{p} / \partial \hat{t}) \in \mathbb{R}^8$.

6 Loss functions

Let $\{\mathbf{x}_n\}_{n=1}^N$ be the data points (with measured displacements $\hat{\mathbf{u}}_n^{\text{data}}$), $\{\mathbf{x}_i^c\}_{i=1}^{N_c}$ collocation points sampled uniformly over the standardized domain box for the wave and gauge residuals, and $\{\mathbf{x}_i^{\text{ic}}\}_{i=1}^{N_i}$ points on the $t = 0$ plane. Each loss is a mean square.

Data loss. Compares the displacement (10) (derived from the predicted potentials) to the measurements:

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_{n=1}^N \left[(\hat{u} - \hat{u}^{\text{data}})_n^2 + (\hat{v} - \hat{v}^{\text{data}})_n^2 + (\hat{w} - \hat{w}^{\text{data}})_n^2 \right]. \quad (18)$$

Wave-residual loss. Sums the squared residual (14) over the four potentials:

$$\mathcal{L}_{\text{wave}} = \frac{1}{N_c} \sum_{i=1}^{N_c} \sum_{q \in \{\hat{\phi}, \hat{\psi}_x, \hat{\psi}_y, \hat{\psi}_z\}} r_q(\mathbf{x}_i^c)^2. \quad (19)$$

Gauge loss. Penalizes departures from $\nabla \cdot \Psi = 0$ via (16):

$$\mathcal{L}_{\text{gauge}} = \frac{1}{N_c} \sum_{i=1}^{N_c} g(\mathbf{x}_i^c)^2. \quad (20)$$

Initial-condition loss. Enforces (17):

$$\mathcal{L}_{\text{ic}} = \frac{1}{N_i} \sum_{i=1}^{N_i} \|\mathbf{r}_{\text{ic}}(\mathbf{x}_i^{\text{ic}})\|^2 = \frac{1}{N_i} \sum_{i=1}^{N_i} \sum_q \left(q^2 + q_{\hat{t}}^2 \right) \Big|_{\hat{t}_0}. \quad (21)$$

Total loss and gradient-norm balancing. The physics terms are grouped as

$$\mathcal{L}_{\text{phys}} = \mathcal{L}_{\text{wave}} + w_g \mathcal{L}_{\text{gauge}} + w_i \mathcal{L}_{\text{ic}}, \quad \mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{phys}}. \quad (22)$$

The physics weight λ is set each epoch so that the physics gradient is a fixed fraction α of the data gradient (preventing the large second-order physics gradients from overwhelming the data fit),

$$\lambda^* = \alpha \frac{\|\nabla_{\theta} \mathcal{L}_{\text{data}}\|}{\|\nabla_{\theta} \mathcal{L}_{\text{phys}}\|}, \quad \lambda \leftarrow \beta \lambda + (1 - \beta) \lambda^*, \quad (23)$$

where θ are the network weights, ∇_{θ} the gradient with respect to them, $\|\cdot\|$ the Euclidean norm, and $\beta = 0.9$ an exponential-moving-average factor. A short data-only warmup ($\lambda = 0$ for the first 12 epochs) precedes the physics phase.

7 Training

Optimizer: Adam, learning rate 2×10^{-3} , gradient-norm clipping at 1, and a plateau learning-rate schedule. All derivatives (first order for the displacement, second order for the wave residual) use automatic differentiation in double precision. The best model is the one with the lowest validation data loss. Data and collocation are drawn in minibatches (data batch 16384).